Vanadis 3D - Atmospheric dispersion model

Vanadis 3D is a three-dimensional numerical model designed for the simulation of atmospheric pollutant transport and dispersion. The model is based on the finite-element method and is intended for problems in which advective transport may dominate over diffusion.

Numerical stabilization is provided by the Directional Residual (DR) method. This approach helps reduce excessive numerical diffusion and improves the behavior of the solution in strongly advection-dominated transport problems.

The computational domain is divided into three-dimensional HEX8 finite elements. For each element, local matrices are constructed to represent transport, diffusion, reaction processes and boundary conditions. Vanadis uses an Element-by-Element strategy, so a conventional global sparse matrix does not need to be assembled.

Finite-element matrix construction is performed on the CPU and can be parallelized using OpenMP. The resulting linear system can be solved either on the CPU or on a GPU using CUDA. The GPU implementation uses an iterative DBCG solver with lightweight diagonal preconditioning.

Two CUDA Element-by-Element execution strategies are available. The first uses atomic operations when different elements contribute to shared global nodes. The second uses graph coloring, which divides the finite elements into independent groups and allows matrix-vector operations to be performed without atomic updates within each color.

Vanadis supports Dirichlet boundary conditions as well as flux-type boundary conditions, including pollutant deposition at the ground surface.

The model performs transient simulations. At every time step, the local finite-element system is constructed and the resulting algebraic problem is solved iteratively.

The current version of Vanadis also supports a nonlinear concentration-dependent reaction coefficient $P(S)$. The nonlinear problem is solved using Picard iteration, with the finite-element system being updated and solved repeatedly within each time step until convergence is reached.

Emission sources are represented through a generalized time-dependent source term $Q(t)$. This allows the source strength to vary during the simulation and makes it possible to represent constant, switched, pulsed or otherwise time-varying emissions.

The overall architecture of Vanadis preserves the local Element-by-Element structure. The CPU is primarily responsible for preparing the physical problem, evaluating model parameters and constructing element matrices, while the GPU can perform the computationally intensive iterative solution of the algebraic system.

Vanadis 3D is therefore a parallel finite-element atmospheric dispersion model for transient pollutant transport, diffusion and reaction, designed to exploit both multi-core CPUs and CUDA-capable GPUs.